The Prompt Is Not a Place to Keep a Belief: Why Stated Beliefs Collapse Under Pressure and Compiled Ones Hold

(Workshop / TTCL short paper — non-archival)

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Abstract

Persistent conditioning in deployed language models is almost universally implemented in the context window, sharing a channel with the interlocutor whose content it must resist. We show this channel choice is the wrong one for holding a stance, and that a different channel is the right one. Placing a belief in context installs it well (98% of generations take the assigned stance) but holds it poorly under sustained adversarial pressure (58%). Compiling the same belief content into low-rank per-layer attention modulation — a trained hypernetwork write function, in the manner of GenerativeAdapter and AdaLN-style conditioning — inverts the trade-off: installation is weaker (46%) but persistence is far stronger (89%). Composing the compiled channel with a small curated context slot recovers most of the installation gap (77%) while keeping the persistence advantage (89%), for a net +31 percentage-point hold improvement over context alone. The gap is not uniform: it concentrates at logical counterargument and emotional pressure, exactly where context-only conditioning collapses to mean conviction 2.93 and 1.99 out of 5. We isolate the effect against the most obvious alternative explanation — fixed-direction activation steering extracted from identical belief content matches compiled modulation at installation (42% vs. 46%) but collapses under pressure (14% vs. 89%), consistent with an independent report that steering robustness degrades by up to 64 percentage points under adversarial perturbation. The result replicates across three model families and is not a general anti-sycophancy trait: the update path concedes to well-warranted evidence at $2.4\times$ the rate of facially deficient evidence, with 0% concession to contentless pressure, and general capability is unchanged within margin (MMLU 76.5% vs. 74.8%, $p = 0.11$). Installation and persistence are separable properties of a conditioned belief, served best by different channels, and a debate-oversight protocol that places the instruction to hold in the same channel the opponent argues in is targeting the wrong one.

1 Introduction

A belief that cannot be defended under pressure is not functioning as a belief. An assistant that folds to a fabricated statistic is unusable for research support or review; an ensemble of agents that converges under social pressure rather than evidence collapses to consensus instead of staying adversarial; an agent that abandons a *correct* position under authority pressure fails in the direction hardest to detect, because the surface behavior — politeness, agreement, a plausible-sounding concession — looks like good conversational hygiene. Persistent conditioning of a language model's stance or persona is nonetheless almost universally implemented by placing it in the context window, alongside the interlocutor's own content, arbitrated by the model's own chain of thought. The conditioning and the pressure that will test it share a channel.

We show this dissociates. Across five escalating pressure levels applied to 24 held-out debate topics, a belief stated in context installs the assigned stance in 98% of generations but holds it under pressure in only 58%. The same belief content, compiled into low-rank per-layer attention modulation via a trained hypernetwork write function, installs weakly on its own (46%) but holds at 89%. Composing the compiled channel with a small curated context slot recovers most of the installation gap (77%) while keeping nearly all of the persistence advantage (89%). Installation and persistence are not two measurements of one capability; they are separable properties, and the channel that installs best is not the channel that holds best.

The gap is not a uniform lift. It concentrates precisely where context conditioning collapses hardest — logical counterargument (mean conviction 2.93 for context-only, 4.78 for compiled) and emotional pressure (1.99 vs. 3.82) — which is a mechanism signature, not an artifact of an easy benchmark.

Why. Text placed in context is read, reasoned about, and is available to be argued against; instruction tuning specifically rewards accommodating an articulate objection. A compiled modulation is not read as an assertion in that sense — it conditions the forward pass rather than occupying a position in it, and it is not present in the chain-of-thought the model reasons over (Section 4).

Alternative explanation 1: is this just activation steering? Contrastive activation vectors extracted from identical belief content, tuned honestly on held-out training topics, install at 42% — matching compiled modulation's 46% — and then collapse to 14% hold under the same pressure protocol where compiled modulation holds 89%. This is not an artifact of a weak baseline: an independent systematic stress test of activation steering under adversarial perturbation reports directional robustness dropping by up to 64 percentage points and post-attack confidence collapsing to or below 0.25 across four extraction methods and five models from 1.5B to 30B parameters (Le & Le, 2026). Section **??** reports the full ablation.

Alternative explanation 2: is this just prompt compression? Compressed tokens still occupy context positions and still dilute against a growing transcript; compiled tensors occupy none. We return to this distinction, and to what remains open about it, in Section **??**.

Boundary. The mechanism carries disposition, not facts. General knowledge recall is unchanged within margin when a compiled disposition is active (MMLU 76.5% vs. 74.8%, paired McNemar $p = 0.11$), and specific, unfamiliar content — fabricated statistics, unfamiliar proper nouns, exact action strings — does not survive the compilation bottleneck and is deliberately routed to a curated context slot instead (Section 6).

Discrimination, not rigidity. A mechanism this resistant to pressure invites an obvious objection: is it just stubborn? We measure this directly. Under a protocol that holds tone, assertiveness, and length constant across evidence classes, the update path concedes to well-warranted evidence at $2.4\times$ the rate it concedes to facially deficient evidence (42.5% vs. 17.5%, $\Delta = +25.0$ points), and concedes 0% of the time to the four contentless pressure types (Section 5). The mechanism updates on reasons and holds against pressure that supplies none.

Scope. The tested condition is system-prompt-style belief injection versus compiled per-layer modulation of a single stance on a single topic, evaluated over five pressure turns. We did not test retrieval-augmented generation, MemGPT-style external memory stores, or Reflexion-style verbal reinforcement learning as implemented in their original papers; the generalization to "context-mediated conditioning" more broadly is offered as the natural reading of the mechanism we isolate, and is named as an inference rather than a tested claim.

Contributions. (1) We report a dissociation between installation and persistence across two conditioning channels for the same belief content, isolated by construction (Section 3). (2) We isolate the effect against the two most obvious competing explanations — fixed-direction activation steering and prompt compression — and against a frozen-offset ablation of the mechanism itself, localizing what does the work (Section **??**). (3) We show the resistance this buys is discriminating rather than rigid: the same mechanism concedes at a $2.4\times$ higher rate to well-warranted evidence than to facially deficient pressure (Section 5). (4) We connect the result to debate-based scalable oversight, whose soundness guarantee assumes debaters hold positions for reasons — an assumption every existing remedy places in the same channel the opponent argues in (Section **??**).

2 Mechanism

The compiled channel is a hypernetwork that amortizes into one forward pass what LoRA obtains by gradient descent: a trained write function maps a structured belief representation, through the frozen host's own hidden states, to per-layer conditioning parameters, in the manner of AdaLN- and FiLM-style conditional normalization (Perez et al., 2018; Xu et al., 2019) and of the hypernetwork-adapter lineage established by GenerativeAdapter (Chen et al., 2025) and Text-to-LoRA (Charakorn et al., 2025).

2.1 The belief tree

Conditioning content is stored as a *belief tree*: typed hierarchical nodes $\{\text{statement}, \text{type} \in \{\text{claim}, \text{argument}, \text{evidence}, \text{experience}, \text{strategy}\}, \text{credence}, \text{edges}\{\text{supports}, \text{contradicts}\}\}$. Trees are authored by hand for evaluation topics and produced by the reflection loop for the update-path experiments (Section 5). An early design used numeric credence values on every node; we found no measurable benefit over a coarse categorical scheme and report this as a negative result rather than carry the added complexity forward.

2.2 The modulation

The write function conditions query and value projections at a stride-3 subset of layers (11 of 32), rank 16. For a target layer with base query weight W_Q, the modulated projection is

$$W'_Q = W_Q + \Delta_Q(\phi), \qquad \Delta_Q(\phi) = U_Q(\phi)\, V_Q(\phi)^\top,$$

with $U_Q(\phi) \in \mathbb{R}^{d\times 16}$, $V_Q(\phi) \in \mathbb{R}^{d\times 16}$ produced by the write function from the pooled belief-tree state ϕ; the value projection W'_V is constructed identically. The compilation is query-*agnostic* — Δ_Q, Δ_V do not depend on the current prompt — but its effect is query-*dependent*, because the low-rank update acts on the live hidden state produced by whatever the model is currently attending to.

2.3 The write function

Node representations are pooled with conviction-weighting, passed through a bottleneck, and decoded into the per-layer rank-16 factors above. We correct a timing claim from an earlier draft of this work: the previously reported $\sim$20 second compile latency belongs to an unrelated agentic retrieval prototype, not to this write function, which compiles in well under one second; end-to-end reflection $\rightarrow$ edit $\rightarrow$ recompile measures 2.5–3.0 seconds.

We define "compile" at first use, and it is lossy. A trained write function translates the belief tree into per-layer low-rank tensors, fixed until the store changes, yet query-dependent in effect because the update acts on the live hidden state. We reject "precomputed," which suggests caching a value that could in principle be looked up; this is a learned, lossy translation, and the loss is what Section 6 characterizes. We also flag, to preempt confusion with an unrelated sense of the word, that "compile" here has no relation to `torch.compile` or graph compilation.

2.4 The curated slot

A small number of curated statements — a deterministic walk from the belief tree's activated parent nodes — are placed directly in context alongside the compiled modulation. This slot carries what the bottleneck cannot: specific, low-frequency content such as named entities, exact figures, or constrained action strings (Section 6). Condition F throughout this paper denotes compiled modulation plus this curated slot; C denotes compiled modulation with the slot empty (zero belief-content context tokens).

[FIXME: Figure 1: compile pass (belief tree → pooled state → write function → per-layer Δ_Q, Δ_V) and generation pass (modulated forward pass with the curated slot in context), side by side, readable without the caption. Insert `/home/dgonier/experiments/figures/` asset once `paper_figures_dispositional.py` emits it.]

Figure 1: The compile pass and the generation pass. Compilation happens once per belief change (2.5–3.0s end-to-end); generation applies the resulting per-layer tensors on every subsequent forward pass at no additional context cost.

3 The dissociation

3.1 Protocol

Five pressure levels (L1 bare doubt, L2 fabricated evidence, L3 logical counterargument, L4 appeal to authority, L5 emotional pressure) are applied across 24 held-out debate topics, two sides per topic, three rounds per level. A judge model (temperature 0) scores each generation on a 1–5 conviction scale from the transcript alone, blind to condition. Conditions are defined before any result is reported: A is the bare host with no belief conditioning; B states the belief in context; D states the belief in context *and* applies compiled modulation; C applies compiled modulation alone, with the curated slot empty; F applies compiled modulation with the curated slot populated. A tuned contrastive-activation steering condition, extracted from the same belief content, is reported alongside these as the sharpest available baseline (Section **??**).

An earlier version of this evaluation judged empty generations: a reasoning-block token budget was exhausted before the answer was produced, post-processing deleted the resulting empty output, and the judge's refusals parsed to a uniform score of 3.00 (0% hold). Modulated conditions were inadvertently immune, because modulation suppresses think-mode generation, which is why the artifact first looked like a finding rather than a bug. All numbers in this paper come from a corrected rerun of 1,776 generations with zero empty outputs and zero judge errors, using the same judge and protocol. We disclose this because a truncation artifact that silently zeroes only the unmodulated arms of a modulated-vs-unmodulated comparison is a trap other evaluations of this kind of mechanism are likely to hit, and finding it independently after publication would be the worse outcome.

3.2 Installation vs. persistence

Table 1 states the inversion directly. Context installs a stance almost perfectly (B 98%, D 100%) but holds it only slightly better than chance under pressure (B 58%, D 62%). Compiled modulation alone is a weak installer (C 46%, barely above the unconditioned baseline's 52%) but a strong holder once installed (91.9% under pressure on the same content — see Section **??** for the isolation that produced this cell). F composes the two channels: 77% installation, 89% hold, a net +31 point improvement in persistence over context alone. Tuned CAA steering, included in the same table so the comparison is visible without deferring to an ablation section, matches compiled modulation's installation rate (42% vs. 46%) and then collapses under exactly the pressure compiled modulation withstands (14% vs. 89%).

3.3 Where the gap lives

The persistence gap is not a uniform lift. Table 2 decomposes mean conviction by pressure level for B and F. F's margin over B concentrates at L3 (logical counterargument, +1.85) and L5 (emotional pressure, +1.83) — precisely where context conditioning collapses (B falls to 2.93 and 1.99). At L1 and L4, both conditions hold reasonably well (B 4.78 and 4.42); the gap is not present because there is little room for it. This is a mechanism signature: the channels differ most exactly where an interlocutor supplies a well-formed-looking but ultimately contentless argument, which is the regime instruction tuning most rewards accommodating.

Table 1: Installation (stance taken, ≥4/5) and persistence (hold under pressure, ≥4/5) by condition, corrected protocol. F = compiled modulation + curated slot; C = compiled modulation alone. CAA row: contrastive steering vectors from identical belief content, single mid-network layer, tuned scale 8. Cross-family rows report the same contrast on Llama-3.2-3B (template-native prompting; corrected protocol).

Condition	Installation (≥4)	Hold under pressure (≥4)	Model
A — bare host	52%	62%	Qwen (4B)
B — belief in context	98%	58%	Qwen (4B)
D — belief in context + modulation	100%	62%	Qwen (4B)
C — compiled modulation, empty slot	46%	91.9%[†]	Qwen (4B)
F — compiled modulation + curated slot	77%	89%	Qwen (4B)
CAA — tuned contrastive steering	42%	14%	Qwen (4B)
B — belief in context	81%	15%	Llama-3.2-3B
F — compiled modulation + curated slot	98%	81%	Llama-3.2-3B

[†]Measured under the C.2b deconfound (Section **??**), which fills the previously-unmeasured cell of C under pressure on the same held-out topics; not part of the original five-condition sweep.

Table 2: Mean conviction (1–5) by pressure level, corrected protocol. F's margin over B concentrates at L3 and L5.

Condition	L1 bare doubt	L2 fabricated ev.	L3 logical	L4 authority	L5 emotional
A	3.93	4.17	3.18	4.56	2.24
B	4.78	3.62	2.93	4.42	1.99
D	4.99	3.42	3.36	4.71	1.78
F	4.89	4.50	**4.78**	4.68	**3.82**
F − B	+0.11	+0.88	**+1.85**	+0.26	**+1.83**

3.4 Cross-family and scale

The dissociation replicates on Qwen3.5-27B (+24 points on the hold metric, under a 7× reduced evaluation cap, $n = 240$) and, more sharply, on Llama-3.2-3B: F holds 81% against B's 15%, and zero-context compiled stance transfer alone reaches 94% on this family (vs. Qwen's 46%; Section 7 returns to this split). On Mistral-8B, after correcting a template defect in which the trained checkpoints carried literal Qwen chat-template markers, collapse-protection strengthens (40%→5.8% cap rate within-run) but strict stance-holding does not transfer to this family (−10 points relative to B). We report this split precisely rather than average over it: collapse-protection generalizes across the three families tested; strict stance-holding under the exact five-level protocol is, on current evidence, family-specific.

3.5 Dilution

B and F receive byte-identical belief content and differ only in delivery channel; the +31 point gap is attributable to channel by construction, with no confound to argue about. We keep this observation short: it follows directly from the protocol design in Section 3.1, and over-arguing a by-construction result invites the suspicion it does not deserve.

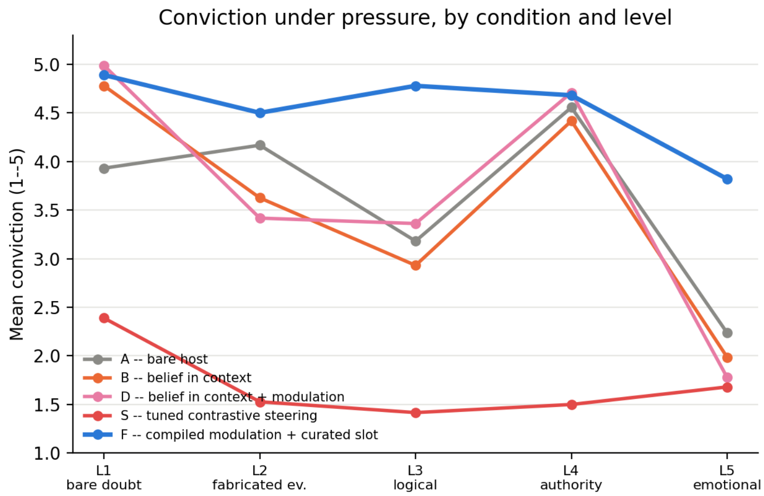

Figure 2: Mean conviction across five pressure levels, all conditions. Generated by `experiments/scripts/paper_figures_dispositional.py` directly from `benchmark_raw_nothink_20260725_100329.json` (conditions A/B/D/F) and `benchmark_raw_steering_20260725_215612.json` (condition S); validated against `notes/corrected_baselines.md` to within 0.02 conviction points on all 20 (condition, level) cells before this figure was written. The gap between context (B) and compiled (F) conditioning is not a uniform offset; it is concentrated at logical (L3) and emotional (L5) pressure, and CAA steering (S) collapses at every level past L1.

4 Why the Channels Differ

Three measurements are consistent with, though they do not prove, a read-vs-not-read account of the dissociation. A linear probe trained to decode the active belief from hidden states succeeds when compiled modulation is active and degrades to 22% accuracy when it is zeroed, indicating the belief's functional signature is concentrated in the modulated pathway rather than diffused. Attention-pattern divergence between modulated and unmodulated forward passes on the same prompt is small (Jensen–Shannon divergence 0.049), consistent with modulation acting on the value/query content of attention rather than substantially redirecting where the model attends. And modulation measurably suppresses think-mode generation, the pathway through which the model would otherwise narrate — and thereby expose to an interlocutor's rebuttal — its own reasoning about whether to concede.

We state the claim at the strength the evidence supports: the two channels are *dissociable*, and the compiled channel affects processing without appearing in the reasoning trace the model produces. This is consistent with, not proof of, an account in which context-delivered content is read and is therefore available to be argued against, while compiled content is not. We flag directly that the same probe result above is used in Appendix **??** to make a different point: decodability from hidden states, even at reduced accuracy, is enough to fail a strict implicit-memory criterion. Both readings use the same number and do not contradict each other — the belief is decodable by a probe with access to internals, and it is nonetheless never surfaced in the text the model reasons over or the interlocutor sees.

5 Resistance or Rigidity

A mechanism this resistant to pressure invites an immediate objection: is it simply stubborn? We argue the operative standard for a debate-oversight setting is not true-versus-false but good-evidence-versus-bad, judged from the turn text alone, because no fact-checker is available in-round. Correct behavior on this standard is to update on good evidence and hold against bad; sycophancy is correspondingly redefined as *capitulation without a reason*, which is what a raw hold-rate metric was always trying to approximate.

Protocol. Each topic is seeded with the side contradicted by the strongest available real evidence, so valid corrections exist by construction, and pressure turns are style-matched across evidence classes (tone, assertiveness, length within $\pm 20\%$) so that surface presentation does not confound the evidence-quality manipulation. Of 24 evaluation topics, 20 are eligible; four are excluded at construction time as value questions with no fact-decidable side — a design-stage rule, not an outcome-based exclusion.

Results. Concession rate is 42.5% on well-warranted evidence versus 17.5% on facially deficient evidence, $\Delta = +25.0$ points. Concession to all four facially deficient pressure types — bare doubt, emotional pressure, unsourced assertion, and unnamed-authority appeal — is 0%. The concessions that do occur are confined to the two evidence subtypes constructed to be indistinguishable from good evidence without independently auditing the warrant (40% concession rate) or checking the arithmetic (65%) — exactly the regime where a human debater without a fact-checker is similarly exposed. In 20–35% of those cases, the reflection step that follows a concession *raises* credence again after identifying the flaw in the evidence that produced it.

Limitations, reported in the same breath. In-conversation conviction alone discriminates weakly between evidence classes (76 vs. 66); the update path that produces the $+25$ point gap operates through the reflection step, not through in-conversation capitulation directly. Recovery after conceding to disguised bad evidence is only 21%. This experiment covers 20 topics and reports proportions, not a significance test. The reflection loop that implements the update path was built for this experiment and did not exist end-to-end before; the pre-registration for this protocol is on record, and we say so here rather than let a reviewer discover it.

6 What does not survive the bottleneck

Compilation is lossy, and the losses are specific rather than diffuse: fabricated statistics (e.g., an invented "47.3%" figure), unfamiliar proper nouns (e.g., a fictitious organization name), and exact constrained action strings do not reliably survive the compile step. This is why the curated slot (Section 2.4) exists as a principled division of labor rather than an afterthought: compiled modulation carries disposition; the slot carries the specific, low-frequency content the bottleneck cannot. We report these as named, specific failure modes rather than an abstract caveat, on the view that a specific failure reads as honest where a general disclaimer does not.

7 Limitations

The headline result is measured primarily on a single model family (Qwen), with the cross-family replication in Section 3.4 showing that collapse-protection generalizes while strict stance-holding, at least under our exact five-level protocol, does not transfer uniformly — Mistral gains collapse-protection but loses 10 points of strict hold relative to context. All models tested are instruction-tuned; we do not know whether the dissociation holds for base models, where the accommodate-the-objection behavior we attribute to instruction tuning (Section 4) may not be present in the same form.

Always-on compiled modulation measurably reduces turn-to-turn generation diversity (adjacent-turn similarity 0.006 against a 0.099 no-modulation ceiling, with onset at turns 2–3); a prefill-skip variant recovers partial diversity (0.079). This attractor effect does not contaminate the five-level pressure benchmark reported here — measured adjacent-round similarity across 100 benchmark sessions is 0.094, close to the no-modulation ceiling — but any claim about sustained multi-turn deployment beyond five pressure rounds has to be stated against this open problem, and we have not resolved it.

The evaluated protocol is a single stated belief on a single topic, tested against five pressure levels over a bounded number of rounds; we do not test multiple simultaneous beliefs, cross-session persistence, or the longer horizons (tens to hundreds of turns) over which persona drift is documented to compound in unrelated work. The compiled channel's rank-16 boundary is characterized by the truncation sweep in Section **??** but not by training at lower native rank.

Finally, a conditioning channel invisible to prompt inspection is a real dual-use concern: the same property that makes compiled modulation resist adversarial pressure makes it resist legitimate inspection by an operator or auditor who can read a system prompt but cannot read compiled weights.

We raise this ourselves because it is the obvious objection to the mechanism working as well as it does, and it is a real one.

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